Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Garland, Jay L Ph.D.		
Project Title:	Biological Water Treatment Technology: Integrated Evaluation of Submerged Bioreactor Technology		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Strayer, Richard (Dynamac Corporation) Levine, Lanfang (Dynamac Corporation) Roberts, Michael (Dynamac Corporation) Hummerick, Mary (Dynamac Corporation) Rector, Tony (Dynamac Corporation)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This Technology Development Project proposes to evaluate a submerged membrane bioreactor (MBR) design for treatment of wastewater for space flight applications. The combination of membrane separation technology and stirred tank bioreactor results in a water treatment system that is compact with high effluent quality, making it an attractive option for ALS water treatment applications. The retention of solids, including bacteria, within the reactor results in higher biomass concentration and concomitant improvement in biodegradation efficiency. The long solids retention time allows for the cultivation of slow-growing organisms (i.e., nitrifying organisms) and causes a general reduction in the growth of new microbial biomass, reducing problems associated with continual accumulation of microbial biomass (i.e., sludge) observed in other reactor designs. Initial trials with the MBR during FY2002 found complete degradation of the organic load (i.e., surfactants) within the wastewater analog with a 12.6 hr hydraulic retention time, indicating that a 12-L reactor could process the simulated wastewater stream (urine, graywater, and atmospheric humidity condensate) from a 2-person crew. However, no nitrification was observed during the 65-day test, and significant abiotic precipitation was problematic. Since the nominal reactor pH was ~8.5, FY2003 activities will determine if pH control will increase nitrification and reduce precipitation. Concurrent studies will involve diluting the urine stream to determine if high ammonia levels are inhibiting nitrification. Follow-up studies will be conducted to determine the level of pH control and influent dilution necessary to achieve optimal organic degradation and nitrification in the single-stage reactor. The first set of tests will determine if nitrification and denitrification processes can be supported in the same reactor by reducing aeration rates, thereby limiting costs associated with supplying oxygen to the system. The second set of tests will evaluate reactor performance and system stability in response to variation in surfactant loading rates simulating temporal fluctuation of crew hygiene inputs. This work will include the addition of the antibiotic amoxicillin to determine the fate and effect of the antibiotic in the reactor. Amoxicillin is a potential contaminant of concern in the system as it is approved for crew use and is excreted in the urine following oral application. Research within the second year will focus on optimizing simultaneous nitrification/dentrification in both gravity-dependent and -independent (i.e., “bubble-less” aeration via membranes) designs, while the final year will concentrate on long-term performance (6-12 month) with actual wastewater streams. A major component of the work within all three years will be analytical efforts to characterize the organic/inorganic chemistry and microbiology of the system. Chemical analyses will be performed to evaluate the kinetics and degradation pathway of major organic constituents (i.e., surfactants), and to quantify the fate and effect of trace organic contaminants (e.g., amoxicillin). Development of new analytical procedures has been, and will continue to be, an important component of this work. Microbiological characterization efforts will involve determining the spatial distribution of community members responsible for nitrification and denitrification, and evaluating the potential microbiological risks associated with the 1) survival and growth of human-associated microorganisms within the system, and 2) development of antibiotic resistant organisms in response to antibiotic addition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	TBD.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)